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DEPARTMENT OF CSE

PROJECT ABSTRACT SUBMISSION

COURSE: ENGINEERING CAPSTONE PROJECT – 1(23IE4053R/23IE4053A) A.Y. 2026 - 2027

TITLE OF THE PROJECT	SmartTour: An Intelligent Tourist Places Guide	
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Abstract

The rapid growth of the tourism industry has significantly increased the demand for intelligent digital platforms that provide travellers with accurate, organized, and easily accessible information about tourist destinations. Traditional methods of collecting travel information from multiple websites and applications are often time-consuming, inconsistent, and inconvenient. To overcome these challenges, the **Tourist Places Guide** has been developed as an adaptive software solution that offers users a centralized platform to discover, explore, and manage information related to tourist attractions. This project was designed as part of the **Adaptive Software Engineering Hackathon**, with the objective of demonstrating the development of scalable, modular, and user-oriented software that can adapt to evolving user requirements.

The Tourist Places Guide provides an interactive and intuitive interface where users can search for tourist destinations by entering the name of a place or location. Once a destination is selected, the application displays comprehensive information, including its historical background, cultural significance, geographical details, and a collection of high-quality images that help users visualize the destination before planning their visit. The project uses a structured dataset of tourist attractions, ensuring that the information presented is accurate, relevant, and organized in a user-friendly format.

One of the major features of the application is the integration of a map service that enables users to locate tourist destinations and understand their geographical position. This feature simplifies navigation and assists travellers in planning efficient routes. Additionally, the system provides recommendations for nearby attractions, allowing users to explore multiple destinations within the same region and maximize their travel experience. By presenting related tourist locations, the application encourages users to discover hidden attractions that they may not have considered during their initial search.

To enhance user convenience, the Tourist Places Guide includes a Favourites module where users can save their preferred destinations for future reference. This personalization feature eliminates the need to search repeatedly for frequently visited locations and contributes to a better overall user experience. The

modular design of the application also ensures easy maintenance and future scalability, allowing developers to integrate additional functionalities without affecting the existing system architecture.

The project follows the core principles of Adaptive Software Engineering by emphasizing flexibility, maintainability, modularity, and continuous improvement. The software architecture is designed to accommodate future enhancements such as AI-based tourist recommendations, multilingual language support, weather forecasting, hotel and transportation booking, user reviews and ratings, travel itinerary generation, offline accessibility, and personalized travel suggestions based on user preferences and browsing history. These potential extensions make the application highly adaptable to changing user expectations and technological advancements.

From a technical perspective, the Tourist Places Guide demonstrates the effective integration of data management, search algorithms, location-based services, and user interface design into a single cohesive application. The project emphasizes efficient data retrieval, responsive system performance, and an intuitive user experience while maintaining software quality through reusable components and modular implementation practices. These characteristics align with the objectives of adaptive software development by ensuring that the application remains reliable, extensible, and easy to maintain over time.

In conclusion, the **Tourist Places Guide** serves as an innovative travel assistance platform that simplifies destination exploration and travel planning by providing historical information, images, navigation support, nearby attraction recommendations, and personalized favourite lists within a unified application. The project successfully demonstrates the application of Adaptive Software Engineering principles to create a practical, scalable, and user-centric software solution capable of evolving with future technological developments and user needs. It highlights the importance of adaptive design in building modern software systems that deliver enhanced functionality, improved usability, and long-term sustainability.

Signature of the Guide

HOD